SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4-**Industry Problem Solving	**Assessment:**	**Assessment Tool-**2 Industry Problem Solving
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	SLOT-D	**Date:**	20-08-2026

Code of Conduct
I SYED MUHAFIZ A(192472282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

Question 1. Software Industry — API Response Time

A software company measures the response time of 100 API requests. The sample mean is 240 ms and the standard deviation is 35 ms. Construct a 95% confidence interval for the average API response time.

Given Information

- Sample size, $n = 100$
- Sample mean, $\bar{x} = 240$ ms
- Sample standard deviation, $s = 35$ ms
- Confidence level = 95% → z-critical value, $z = 1.96$

Step 1: Formula

Since the sample size is large ($n \geq 30$), the standard normal (z) distribution is used to construct the confidence interval for the population mean:

$$CI = \bar{x} \pm z \times (s / \sqrt{n})$$

Step 2: Compute the Standard Error

Standard Error, $SE = s / \sqrt{n} = 35 / \sqrt{100} = 35 / 10 = 3.5$ ms

Step 3: Compute the Margin of Error

Margin of Error, $ME = z \times SE = 1.96 \times 3.5 = 6.86$ ms

Step 4: Compute the Confidence Interval

Component	Value
Lower Limit	$240 - 6.86 = 233.14$ ms
Upper Limit	$240 + 6.86 = 246.86$ ms

Result: The 95% confidence interval for the average API response time is (233.14 ms, 246.86 ms).

Interpretation

We are 95% confident that the true average response time of the API, across all requests, lies between 233.14 ms and 246.86 ms. This means that if the sampling process were repeated many times, approximately 95% of the intervals constructed in this way would contain the true population mean response time. Since the interval is fairly narrow, the company can be reasonably confident that typical response times fall within this range, which is useful for setting service-level agreements (SLAs) and performance benchmarks.

Question 2. E-commerce — Conversion Rate

An e-commerce platform finds that 420 out of 600 randomly selected visitors purchased a product. Construct a 95% confidence interval for the true conversion rate.

Given Information

- Sample size, $n = 600$
- Number of successes (purchases), $x = 420$
- Confidence level = 95% → z-critical value, $z = 1.96$

Step 1: Compute the Sample Proportion

$\hat{p} = x / n = 420 / 600 = 0.70$ (70%)

Step 2: Formula

The confidence interval for a population proportion is given by:

$$CI = \hat{p} \pm z \times \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

Step 3: Compute the Standard Error

$SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.70 \times 0.30) / 600]} = \sqrt{[0.21 / 600]} = \sqrt{0.00035} = 0.01871$

Step 4: Compute the Margin of Error

$ME = z \times SE = 1.96 \times 0.01871 = 0.03667$ ($\approx$ 3.67%)

Step 5: Compute the Confidence Interval

Component	Value
Lower Limit	$0.70 - 0.0367 = 0.6633$ (66.33%)
Upper Limit	$0.70 + 0.0367 = 0.7367$ (73.67%)

Result: The 95% confidence interval for the true conversion rate is (0.6633, 0.7367), i.e. (66.33%, 73.67%).

Interpretation

We are 95% confident that the true proportion of visitors who purchase a product lies between 66.33% and 73.67%. This range gives the e-commerce platform a realistic estimate of its overall conversion performance and can be used to evaluate marketing campaigns, set revenue forecasts, and compare performance against industry benchmarks. Because the sample size is large, the interval is reasonably tight, indicating a fairly precise estimate of the true conversion rate.

Question 3. Telecom — Average Monthly Data Consumption

A telecom company wants to estimate the average monthly data consumption of its users. A sample of 225 users has a mean consumption of 14.8 GB and a standard deviation of 4.5 GB. Construct a 99% confidence interval for the population mean.

Given Information

- Sample size, $n = 225$
- Sample mean, $\bar{x} = 14.8$ GB
- Sample standard deviation, $s = 4.5$ GB
- Confidence level = 99% $\rightarrow$ z-critical value, $z = 2.576$

Step 1: Formula

$$CI = \bar{x} \pm z \times (s / \sqrt{n})$$

Step 2: Compute the Standard Error

$SE = s / \sqrt{n} = 4.5 / \sqrt{225} = 4.5 / 15 = 0.30$ GB

Step 3: Compute the Margin of Error

$ME = z \times SE = 2.576 \times 0.30 = 0.7728$ GB

Step 4: Compute the Confidence Interval

Component	Value
Lower Limit	$14.8 - 0.7728 = 14.0272$ GB
Upper Limit	$14.8 + 0.7728 = 15.5728$ GB

Result: The 99% confidence interval for the average monthly data consumption is (14.03 GB, 15.57 GB) (rounded to 2 decimal places).

Interpretation

We are 99% confident that the true average monthly data consumption of all users lies between 14.03 GB and 15.57 GB. Because a 99% confidence level requires a larger critical value than a 95% interval, this interval is wider than it would be at a lower confidence level, reflecting the higher certainty demanded. This estimate helps the telecom company plan network capacity, design data plans, and forecast bandwidth requirements more accurately.

Question 4. Manufacturing — Component Diameter (Hypothesis Testing using CI)

A machine historically produces components with a mean diameter of 50 mm. A sample of 36 components has a mean diameter of 50.8 mm and a standard deviation of 2.4 mm. Using a 95% confidence interval, determine whether 50 mm is a plausible population mean.

Given Information

- Sample size, $n = 36$
- Sample mean, $\bar{x} = 50.8$ mm
- Sample standard deviation, $s = 2.4$ mm
- Claimed/historical mean, $\mu_0 = 50$ mm
- Confidence level = 95% → z-critical value, $z = 1.96$

Step 1: Formula

$$CI = \bar{x} \pm z \times (s / \sqrt{n})$$

Step 2: Compute the Standard Error

$SE = s / \sqrt{n} = 2.4 / \sqrt{36} = 2.4 / 6 = 0.40$ mm

Step 3: Compute the Margin of Error

$ME = z \times SE = 1.96 \times 0.40 = 0.784$ mm

Step 4: Compute the Confidence Interval

Component	Value
Lower Limit	$50.8 - 0.784 = 50.016$ mm
Upper Limit	$50.8 + 0.784 = 51.584$ mm

Result: The 95% confidence interval for the mean diameter is (50.016 mm, 51.584 mm).

Step 5: Hypothesis Decision Using the Confidence Interval

Hypotheses:

- H_0 (Null Hypothesis): $\mu = 50$ mm (the machine is still producing components at the historical mean)
- H_1 (Alternative Hypothesis): $\mu \neq 50$ mm (the machine's mean diameter has shifted)

Decision rule: If the hypothesized value μ_0 falls within the confidence interval, it is a plausible value for the population mean and H_0 is not rejected. If μ_0 falls outside the interval, it is not plausible and H_0 is rejected.

Here, $\mu_0 = 50$ mm is compared against the interval (50.016 mm, 51.584 mm). Since 50 mm is less than the lower limit of 50.016 mm, it does NOT fall within the confidence interval.

Result: Since 50 mm lies outside the 95% confidence interval, it is NOT a plausible value for the population mean. We reject H_0 and conclude that the machine's average diameter has significantly shifted away from 50 mm.

Interpretation

The evidence suggests that the machine is no longer producing components with the historical mean diameter of 50 mm; the true mean now appears to be somewhat higher, consistent with the observed sample mean of 50.8 mm. This could indicate tool wear, calibration drift, or another systematic change in the manufacturing process. The company should investigate and recalibrate the machine to bring production back in line with specifications, since even a small shift in diameter can affect part quality and downstream assembly.

Question 5. Banking — Loan Processing Time (Hypothesis Testing using CI)

A bank claims that its average loan-processing time is 5 days. A sample of 49 applications has an average processing time of 5.6 days with a standard deviation of 1.4 days. Construct a 95% confidence interval and determine whether the company's claim is reasonable.

Given Information

- Sample size, $n = 49$
- Sample mean, $\bar{x} = 5.6$ days
- Sample standard deviation, $s = 1.4$ days
- Claimed mean, $\mu_0 = 5$ days
- Confidence level = 95% → z-critical value, $z = 1.96$

Step 1: Formula

$$CI = \bar{x} \pm z \times (s / \sqrt{n})$$

Step 2: Compute the Standard Error

$SE = s / \sqrt{n} = 1.4 / \sqrt{49} = 1.4 / 7 = 0.20$ days

Step 3: Compute the Margin of Error

$ME = z \times SE = 1.96 \times 0.20 = 0.392$ days

Step 4: Compute the Confidence Interval

Component	Value
Lower Limit	$5.6 - 0.392 = 5.208$ days
Upper Limit	$5.6 + 0.392 = 5.992$ days

Result: The 95% confidence interval for the mean loan-processing time is (5.208 days, 5.992 days).

Step 5: Hypothesis Decision Using the Confidence Interval

Hypotheses:

- H_0 (Null Hypothesis): $\mu = 5$ days (the bank's claim is correct)
- H_1 (Alternative Hypothesis): $\mu \neq 5$ days (the true average processing time differs from the claim)

The claimed value $\mu_0 = 5$ days is compared against the interval (5.208 days, 5.992 days). Since 5 days is less than the lower limit of 5.208 days, it does NOT fall within the confidence interval.

> **Result: Since 5 days lies outside the 95% confidence interval, the bank's claim is NOT reasonable. We reject H_0 and conclude that the true average loan-processing time is significantly greater than 5 days.**

Interpretation

The sample evidence indicates that loan applications actually take longer to process than the bank advertises, with the true average likely lying between about 5.2 and 6.0 days. This gap between the claimed and actual processing time could affect customer satisfaction and trust. The bank may need to revise its stated turnaround time or investigate ways to streamline its loan-processing workflow to meet its original 5-day claim.

Summary of Results

Q.	Scenario	Confidence Level	Confidence Interval	Conclusion
1	API Response Time	95%	(233.14, 246.86) ms	Estimate constructed
2	Conversion Rate	95%	(66.33%, 73.67%)	Estimate constructed
3	Monthly Data Usage	99%	(14.03, 15.57) GB	Estimate constructed
4	Component Diameter	95%	(50.016, 51.584) mm	50 mm not plausible — reject H_0
5	Loan Processing Time	95%	(5.208, 5.992) days	5 days not plausible — reject H_0